

Week 2: Dimensionless Units

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1 Kinetic Energy Units Derivation

$$K = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{dr}{dt}\right)^2$$

$$K = \frac{1}{2}m\left(\frac{d\tilde{r}}{d\tilde{t}} \cdot \frac{\sigma}{(m\sigma^2/\epsilon)^{1/2}}\right)^2$$

$$K = \frac{1}{2}\tilde{v}^2 \cdot \epsilon$$

Thus, we can write Kinetic Energy in units of ϵ , giving the dimensionless value as:

$$\tilde{K} = \frac{1}{2}\tilde{v}^2$$

2 Force Units Derivation

$$F = m\frac{d^2x_i}{dt^2} = m \cdot \frac{\sigma}{(m\sigma^2/\epsilon)} \frac{d^2\tilde{x}_i}{d\tilde{t}^2}$$

$$F = \frac{\epsilon}{\sigma} \cdot \frac{d^2\tilde{x}_i}{d\tilde{t}^2}$$

Thus, we can write Force in units of $\frac{\epsilon}{\sigma}$, giving the dimensionless value as:

$$\tilde{F} = \frac{d^2\tilde{x}_i}{d\tilde{t}^2}$$